5 Related Literature Review

The literature on pumped-storage hydropower addresses several related questions: whose value is assessed and which capacities are varied, how uncertainty about future conditions is represented, which markets provide revenue, and how the resulting planning problems are solved. This chapter reviews these questions in separate thematic sections, in which the studies are cited, compared, contrasted and critiqued, and connected to the present study. Section 5.1 defines the scope of the review. Section 5.2 examines valuation perspectives and plant capacities, Section 5.3 the representation of uncertainty and price behaviour, and Section 5.4 market participation. Section 5.5 then reviews the solution methods applied to these problems and the physical detail they allow. Finally, Section 5.6 draws the findings together, identifies the research gap and positions the present study.

5.1 Scope and Selection of Literature

The review focuses on operational scheduling and economic valuation of pumped-storage hydropower. Particular attention is given to studies that connect short-term operating opportunities with longer-term reservoir management. Studies of conventional reservoir hydropower are included where they offer relevant insights into water values, uncertainty or reserve markets, since a reservoir plant faces the same storage decision without the pumping option. System-planning and hybrid-energy studies provide context for evaluating storage under different objectives, and studies of electricity-price behaviour are included because the value of storage depends on how prices develop over time.

TODO (authors): Describe the search process you actually used: databases (e.g. Google Scholar, Scopus), search terms, inclusion criteria (peer-reviewed, relevance of title and abstract), and whether further studies were found by following reference lists (backward snowballing). Three to five sentences are enough. The search log (`literature/search-log.md`) is still empty.

The chapter provides a narrative synthesis of the selected studies. Each section addresses one dimension of the literature and compares the studies along that dimension only. Individual studies may therefore appear in several sections, but each discussion concerns the aspect relevant to that section. Table 3 in Section 5.6 summarises the studies across all dimensions.

5.2 Valuation of Pumped Storage and Its Capacities

The perspective adopted in a study determines what is meant by the value of pumped storage, and the plant characteristics it varies determine which part of that value it can explain. This section first examines valuation perspectives and then the capacities that the studies allow to change.

5.2.1 Valuation Perspective

System-oriented studies assess the contribution of storage to the wider electricity system, whereas owner-oriented studies assess the economic benefit available to an individual producer. These objectives need to be distinguished when comparing conclusions about the attractiveness of storage.

System-oriented studies relate storage value to the surrounding electricity system. Helseth et al. (2013) examine pumped storage within a hydrothermal system that also includes wind generation and transmission constraints. Helseth et al. (2018) assess an expansion of the 200 MW Duge pumped-storage plant by up to 2000 MW through changes in expected socio-economic surplus. Both studies consider effects beyond the income of an individual plant. A favourable system-level assessment therefore does not establish the corresponding private return, since benefits can also accrue to consumers and other producers. The two can even diverge: the price differences that pumped storage trades on shrink as more storage is built (Dallinger et al., 2019), and Henden et al. (2016) find that a large expansion of pumped storage in Norway leaves its owners with little income.

Owner-oriented studies instead evaluate the income associated with operating a particular asset. Ak et al. (2019) report revenue gains from operating an existing hydropower cascade in pumped-storage mode, with the gains varying between the investigated price years. Similarly, Alic et al. (2023) assess the economic benefit of adding a 38 MW pump to a Norwegian cascade. These studies address the producer's economic opportunities, although their findings remain specific to the assets and conditions investigated. Increased operating revenue alone also does not establish investment profitability, which requires the associated costs to be considered.

Within an owner perspective, the treatment of market power introduces a further distinction. Abgottspon and Andersson (2013) consider a price-making producer whose trading decisions can influence market outcomes. In contrast, Helseth et al. (2016) consider a price-taking producer facing externally determined prices. This difference affects the interpretation of flexibility: a price taker responds to available price differences, while a price maker also considers the price effects of its own actions.

An owner-oriented valuation can also extend from operating income to an investment assessment. Bore and Brensholm (2026) use additional hydropower revenues to estimate the maximum investment cost that a capacity upgrade can support. This distinguishes an assessment of operating benefits from a valuation that also specifies an economic lifetime, discount rate and operating-cost assumptions. System benefits, operating income and investment profitability are thus different measures of value, and a conclusion about the economic value of storage must be read against the perspective of the study it is drawn from.

5.2.2 Pumping, Generating and Reservoir Capacities

Capacity studies differ in which plant characteristics are fixed and which are allowed to change. Optimising the operation of an existing plant, comparing specified upgrades and

optimising the plant's dimensions address different questions. Distinguishing these tasks is essential when interpreting conclusions about the value of additional capacity.

Studies of specified assets primarily establish how the available flexibility can be used. Naval and Yusta (2025) optimise the operation of a given hybrid-energy configuration, whereas Ak et al. (2019) compare an existing cascade with operation in pumped-storage mode. Alic et al. (2023) and Schüller et al. (2024) similarly evaluate a single pump added to a Norwegian watercourse. These comparisons demonstrate the value of introducing a pumping option under the investigated conditions. They do not establish how that value changes when pumping power, generating power and reservoir capacity are varied independently or jointly.

Comparisons of expansion alternatives address capacity changes more directly. Helseth et al. (2018) assess transmission and pumped-storage expansions, and find that a detailed representation of the watercourse lowers the marginal value of additional pumped-storage capacity. The paper does not state whether pump and turbine capacity are expanded separately. Helseth et al. (2017) compare two generator configurations for a hydropower producer that sells reserve capacity. Both evaluate changes in available flexibility, but the characteristics varied and the valuation settings differ. These studies provide precedents for comparing alternatives without establishing a common ranking of pump, turbine and reservoir capacities for pumped-storage plants.

Bore and Brensholm (2026) examine generating-capacity upgrades of 30, 60 and 90 MW in a conventional hydropower case. They find diminishing incremental value per unit of additional capacity, with reserve-capacity revenues contributing most of the gains. This is a close precedent for assessing capacity changes, but the alternatives concern generating-power upgrades. They do not compare combinations of pumping power, generating power and reservoir size in a pumped-storage plant.

Design optimisation brings selected dimensions into the decision problem itself. Zisos et al. (2023) optimise reservoir dimensions and photovoltaic capacity in a hybrid island-energy system while holding pumping and generating power fixed at 6 MW. Their optimised reservoir is much smaller than the preliminary design, and uncertainty produces a range of storage sizes and economic outcomes. This shows that storage sizing can matter for economic performance. However, the plant is operated by a fixed rule without electricity prices, and fixing pump and turbine power leaves their joint sizing with the reservoir outside the examined design choices.

The reviewed studies therefore provide evidence on operating given assets, introducing pumping, comparing expansions and selecting reservoir dimensions. They differ in which capacities are varied and which remain fixed. This distinction matters because an additional unit of one capacity may have limited value when another capacity is binding. A larger pump, for example, offers limited additional storage opportunity when the reservoir is already full. None of the reviewed studies varies pumping power, generating power and reservoir capacity jointly for a plant that trades in electricity markets.

5.3 Representation of Uncertainty

The value of stored water depends on prices and inflows that are unknown when operating decisions are made. The reviewed studies differ in which inputs they treat as uncertain, how these inputs are described, and when the uncertainty is assumed to be resolved. All three choices affect what a reported storage value represents. In addition, the value of storage depends on how prices behave over time, which few of the studies examine.

5.3.1 Sources of Uncertainty

The sources of uncertainty follow the question a study addresses. Owner-oriented studies concentrate on inflows and energy prices (Alic et al., 2023; Bore and Brensholm, 2026), and Helseth et al. (2016) additionally treat reserve-capacity prices as uncertain. System and hybrid-energy studies emphasise the renewable resource instead: Helseth et al. (2013) include wind power alongside inflow, Naval and Yusta (2025) consider wind generation, and Zisos et al. (2023) consider wind, demand and the wind-to-power conversion. Helseth et al. (2018) combine weather uncertainty with exogenous price uncertainty. Results obtained under one set of uncertain inputs therefore cannot simply be transferred to another. For a price-taking owner, price uncertainty is the most direct driver of the value of shifting energy in time, yet it receives no attention in the hybrid-system studies, where prices play a subordinate role.

5.3.2 Describing Uncertainty over Time

The description of the uncertain inputs also varies. Ak et al. (2019) evaluate annual revenues for historical inflows and a set of electricity-price years. This reveals how revenue gains vary between price years, but each year is evaluated on its own, so the results do not show how an operator should act without knowing which year will occur. Naval and Yusta (2025) use chance constraints, which require operating targets to be met with a specified probability. Such a formulation addresses the reliability of a schedule under wind uncertainty; it does not describe how decisions adapt as outcomes are observed. Other studies describe uncertainty as a process that develops between decision stages. Helseth et al. (2017) and Alic et al. (2023) use weekly Markov states, and Bore and Brensholm (2026) construct a Markov chain from clustered historical data. A Markov description retains the dependence between consecutive weeks. This property matters for storage decisions, because the benefit of holding water depends on how long high- or low-price conditions are expected to last.

The dependence between the uncertain inputs is a further choice. Kleiven et al. (2023) find that ignoring the negative correlation between Norwegian prices and inflow lowers water values and changes reservoir operation, but has little effect on expected revenue. Pérez-Díaz et al. (2020) find the correlation between energy and reserve prices too weak to affect water values for a Spanish plant. Such findings depend on the data and the plant, however, and say little about whether the same holds when prices also drive pumping.

A further distinction concerns when information becomes available to the decision maker. In the weekly-stage studies of Alic et al. (2023), Helseth et al. (2016) and Helseth et al. (2017), new information arrives every week and later decisions respond to it. Helseth et al. (2018) also revisit decisions every week, but treat the uncertainty beyond the current week as

resolved in a single step. The two descriptions imply different degrees of foresight about the more distant future, even when the evaluation period is the same. The length of the horizon alone therefore says little about how realistically a study represents decision-making under uncertainty.

5.3.3 Information within the Week

Uncertainty within a decision stage is a separate issue from uncertainty between stages. Bore and Brensholm (2026) assume that prices and inflow for the current week are known when its decisions are made. Their framework thus combines uncertainty between weeks with perfect foresight within a week, which allows short-term operation to be timed better than is possible in practice. Abgottspon and Andersson (2014) examine this simplification directly by comparing known and uncertain descriptions of operation within the week for a Swiss pumped-storage plant. Treating the week as known overvalues water, but describing it as uncertain improves expected profit only when weekly secondary reserves are included. This suggests that within-week foresight is a less restrictive assumption for energy trading than it first appears, but not when reserves are sold. The conclusion rests on a single plant whose capacities are not reported.

5.3.4 Price Volatility and Persistence

A statement that a study "includes uncertainty" is therefore not sufficient to characterise it. For a storage owner trading on price differences, the representation of price behaviour over time is likely to be at least as important as the number of uncertain inputs. Two properties of this behaviour are central: volatility, how much prices vary around their expected pattern, and persistence, how long high- or low-price conditions last (Section ??). Haldrup and Nielsen (2006) find that Nordic day-ahead prices are highly persistent, most of all in the Norwegian price areas, which they link to the slow variation of reservoirs and inflow. A weekly Markov description captures such persistence only through its transition probabilities, and Mo et al. (2001) note that it therefore forgets deviations from the expected price level quickly.

Studies that relate price behaviour to storage value do so with known prices. Ak et al. (2019) find the largest gain from pumping in the price year with the largest variation within the day, although the relation is not one-to-one. Sioshansi et al. (2009) find that most of the arbitrage value in the US PJM market is earned by the first hours of storage, and Bradbury et al. (2014) find that higher volatility raises the return on storage but is not the main driver of the most profitable ratio of storage to power. Löhndorf and Wozabal (2023) report that coordinating day-ahead and intraday trading is worth about an order of magnitude more for a pumped-storage plant than for a reservoir plant without a pump. None of these studies varies volatility and persistence separately or relates them to the value of pump, turbine and reservoir capacity under uncertainty. Since pumping and generating power earn from short price differences, whereas reservoir capacity earns from differences that last over weeks or seasons, the two properties may favour different capacities.

5.4 Electricity Market Participation

Market participation determines the revenue opportunities available to a storage owner. Energy-market participation rewards electricity sales relative to the cost of purchases for pumping. Other products can remunerate the availability of operating flexibility or its actual use. Studies must therefore be compared by the products and settlements included, since these define what their reported revenues cover. The reported gains are not directly comparable either, as market design, price levels and plant characteristics differ between the studies.

Energy markets

Energy-centred studies provide a baseline for assessing the value of pumping. Ak et al. (2019) examine revenue gains associated with electricity-price variations and find that pumped-storage operation raises net revenue by about 6% on average, while Alic et al. (2023) evaluate price-based operation under environmental constraints. Their focus differs from studies that include additional trading opportunities. For example, Abgottspon and Andersson (2013) consider both day-ahead and forward trading. Forward positions extend the producer's commercial opportunities, but they serve a different purpose from payments for maintaining reserve availability.

Reserve capacity

Reserve-capacity participation creates an additional use for the same operating flexibility. Helseth et al. (2016) coordinate sales into separate, sequentially cleared energy and capacity markets and examine the implications for schedules and water values. Abgottspon and Andersson (2014) also include ancillary services in pumped-storage scheduling. In related conventional-hydropower work, Helseth et al. (2017) find that reserve sales change operation and increase operating profit by 7 to 11%, and Pérez-Díaz et al. (2020) report a gain of about 4.5% for a Spanish plant selling automatic frequency restoration reserves. These studies show that reserve participation can affect the allocation of resources as well as provide an additional source of revenue. For pumped storage, the pump widens this flexibility: Dallinger et al. (2019) show that pump-turbines let a plant offer downward reserve without generating, so its reserve provision no longer depends on natural inflow.

Reserve activation

The treatment of activation further distinguishes market representations. Chazarra et al. (2017) include regulation energy associated with upward and downward reserve use, alongside energy and reserve scheduling, and find that reserve capacity is the main source of income for a closed-loop plant. Accounting for regulation energy connects reserve provision to actual electricity consumption or generation. By comparison, an assessment limited to availability payments does not capture all the associated energy settlements and changes in stored water.

Broader market coverage is considered by Bore and Brensholm (2026), who include day-ahead and intraday energy alongside FCR-N, FCR-D and upward and downward aFRR capacity. Actual reserve activation is excluded. Their results indicate that capacity-market participation affects reservoir management and strengthens revenues in periods with low energy prices.

Compared with the regulation-energy treatment in Chazarra et al. (2017), this example highlights that the number of included products and the completeness of their delivery representation are separate dimensions of market coverage.

Consequently, studies of market participation need to distinguish energy trading, reserve-capacity commitments and reserve activation. Revenues from these activities interact because they depend on shared operating capability. Findings from different market combinations cannot be compared solely through their total income, particularly when their treatment of activation and settlement differs. Most of the multi-market evidence also concerns conventional reservoir hydropower. For pumped storage, where the pump offers an additional means of providing reserves, the long-term effect of reserve participation is less well documented.

5.5 Solution Methods

The preceding sections show that the valuation of pumped storage can combine a long planning horizon, uncertain inputs, physical detail and several markets. No solution method accommodates all of these to the same degree, and the physical detail a study can retain depends on the method it uses. This section reviews the methods applied in the literature, whose principles are introduced in Sections ??, ?? and ??, and compares what each permits and requires.

5.5.1 Deterministic and Scenario-Based Optimisation

The most direct approach formulates the scheduling problem as a single optimisation problem with given inputs. Ak et al. (2019) use nonlinear programming together with monthly reservoir operating rules, and solve the problem separately for each price year. Chazarra et al. (2017) formulate a mixed-integer model for short-term scheduling, which allows the discrete operating modes of the plant to be represented. This includes hydraulic short-circuit operation, in which pumping and generation take place simultaneously, and which they find can greatly increase the income of the plant. Naval and Yusta (2025) extend a mixed-integer model with chance constraints to account for wind uncertainty in hourly operation over a year.

These formulations place few restrictions on the plant description, which explains their use where nonlinear or discrete characteristics are central. Their limitation lies in the treatment of time and information. The schedule is optimised for the inputs supplied, and the long-term use of the reservoir is governed by prescribed rules or targets, as in Ak et al. (2019). The methods are therefore well suited to short horizons or to evaluating given conditions, but they do not by themselves provide the value of stored water under uncertainty.

5.5.2 Two-Stage Stochastic Programming and Rolling Horizons

Two-stage stochastic programming introduces uncertainty while keeping a single optimisation problem. Helseth et al. (2018) embed two-stage stochastic linear programmes in a rolling-horizon simulator: each weekly decision is optimised against a set of scenarios for

the future, and the problem is solved again as the simulation advances. The authors choose this approach to handle many reservoirs without aggregation and to use historical data directly. Compared with the deterministic formulations above, the weekly decisions are informed by a range of possible futures.

The approach has two limitations. First, the scenarios are treated as fully known after the first-stage decision, so the future part of each problem is operated with foresight. Second, the computational effort grows with the number and length of the scenarios. Helseth et al. (2018) accordingly reduce the number of scenarios and use a coarser time resolution along them to limit computation time, and still report run times of several days. A rolling horizon thus offers flexibility in the system description at the expense of a coarser representation of how information is revealed.

5.5.3 Stochastic Dynamic Programming

Stochastic dynamic programming (SDP) divides the problem into stages and computes the expected future value for a discrete set of storage levels. Alic et al. (2023) apply SDP to a one-year problem with weekly stages, uncertain inflows and prices, and environmental constraints that depend on the reservoir level, and show that these constraints alter the optimal production strategy. Helseth et al. (2017) combine SDP with weekly mixed-integer linear problems to retain non-convex operating characteristics, and find that a linear version of their model overestimates the profit from reserve sales. Abgottspon and Andersson (2014) use SDP between stages together with detailed subproblems within each stage.

The common strength of these applications is that SDP imposes no convexity requirement on the stage problems, so detailed plant descriptions can be kept while decisions still adapt to new information every stage. The weakness is the discretisation of the state space: the number of state combinations grows rapidly with the number of reservoirs and other state variables. The reviewed SDP applications accordingly concern individual plants or small cascades. The repeated solution of stage problems for every storage level and uncertainty state also becomes demanding when the stage problems themselves are detailed.

5.5.4 Stochastic Dual Dynamic Programming

Stochastic dual dynamic programming (SDDP), introduced by Pereira and Pinto (1991), avoids discretising the state space by approximating the future value function with cutting planes that are refined iteratively. This allows multistage problems with several reservoirs to be solved, and the method has become the standard approach for large-scale hydropower scheduling under uncertainty (Füllner and Rebennack, 2025). Helseth et al. (2013) apply it to a hydrothermal system that includes pumped storage and wind power, with linearised transmission constraints, approximate start-up costs and a fine time resolution, at the expense of a considerable computational burden. They report that the approximated start-up costs and the transmission constraints clearly affect the results.

The method relies on the structure of the problem. The convergence analysis of Shapiro (2011) concerns multistage linear stochastic programmes, and the cutting-plane argument

does not extend directly to non-convex stage problems. Price uncertainty requires particular care, since uncertain prices enter the objective. Helseth et al. (2016) handle this by combining SDP for the price process with SDDP for the reservoirs, which allows inflows, energy prices and reserve-capacity prices to be uncertain. They also note that the required linear relationships can introduce errors in the solution. This is the counterpart of the SDP trade-off: SDDP scales to larger systems and finer resolution, but restricts the physical description of the plant.

Extending the method

Several studies seek to relax this restriction. Hjelmeland et al. (2019) include head-dependent, non-convex generation when energy and reserve capacity are co-optimised, and strengthen the cuts to approximate it. Helseth (2024) represent a detailed hydropower system within an aggregated model through feasibility cuts computed before the SDDP run, building on Helseth and Mo (2023). Both report better solutions together with increased computation time. Such extensions narrow the gap between the two dynamic-programming approaches without removing the underlying trade-off, and none of them concerns pumped storage.

Applications to individual owners

Applications of SDDP to an individual owner's valuation remain concentrated on conventional reservoir hydropower. Bore and Brensholm (2026) use SDDP with a Markov chain for inflow and prices to value generating-capacity upgrades across several markets. For pumped storage, Abgottspon and Andersson (2016) use SDDP in a multi-horizon model that links seasonal reservoirs with small daily reservoirs, but in a deterministic case, and Schüller et al. (2024) use water values from an SDDP-based model to value a pumping investment in the energy market. Owner-oriented pumped-storage studies that include reserve markets have relied on SDP or deterministic optimisation.

Physical detail and the choice of method

The appropriate level of physical detail thus depends on the purpose of the model. Short-term scheduling studies can afford discrete decisions because their horizon is short, whereas studies of seasonal storage simplify the plant to keep the problem tractable. The reviewed studies document that simplifications affect the level of estimated income. They do not examine whether simplifications also change the ranking of alternative plant configurations, which is the relevant concern when configurations are compared.

The methods thus form a progression in which each step gains in one respect and gives up in another. Deterministic and scenario-based formulations allow the most detailed plant description but do not yield an operating policy under uncertainty. Rolling two-stage formulations add uncertainty at the expense of its sequential resolution. SDP represents sequential decisions with full physical detail for small systems, and SDDP extends sequential decision-making to larger problems and finer resolution under linearity requirements. The choice of method consequently determines which of the features discussed in Sections 5.2–5.4 a study is able to represent.

5.6 Research Gap and Positioning of This Study

Table 3 summarises the reviewed studies along the dimensions discussed above. The literature establishes several relevant strands: economic valuation from different perspectives, alternative representations of uncertainty and of the plant, participation in multiple markets, comparison of plant configurations, and solution methods with different capabilities. These strands address complementary aspects of the value of pumped-storage flexibility.

The closest studies cover different parts of the capacity-valuation question. Abgottspon and Andersson (2014) connect pumped-storage scheduling with ancillary-service opportunities, and Helseth et al. (2016) connect uncertain market conditions with water values. Helseth et al. (2017) additionally compare hydropower configurations, while Zisos et al. (2023) investigate reservoir sizing. These contributions provide a basis for analysing capacity interactions, but their different assets, objectives and design choices do not establish a common assessment of pump, turbine and reservoir capacities for a pumped-storage owner.

The thesis by Bore and Brensholm (2026) narrows this research need further by directly connecting stochastic multi-market operation to the value of generating-capacity upgrades. Consequently, the combination of uncertainty, market participation and capacity valuation cannot itself be treated as an unaddressed question. The remaining gap concerns the joint contribution of pumping power, generating power and reservoir capacity in a pumped-storage setting, together with the influence of different price dynamics, which the reviewed studies have examined only with known prices. The review also indicates that owner-oriented pumped-storage studies with reserve markets have so far relied on methods that either omit sequential uncertainty or are confined to small state spaces.

Against this background, the present study asks how pumping power, generating power and reservoir capacity, individually and in combination, affect the owner's long-term operating value across electricity markets, and how the relative value of these characteristics changes with price volatility and persistence. The study adopts the perspective of a price-taking owner and compares predefined plant configurations under common market and resource conditions. In line with the reviewed applications to reservoir hydropower, the problem is solved with SDDP, using weekly stages and uncertain inflow and day-ahead prices. The last row of Table 3 places the study alongside the reviewed literature. The contribution sought is a comparative understanding of capacity interactions under uncertainty; it does not depend on claiming that stochastic scheduling, multi-market participation or storage sizing is individually new.

Table 3: Reviewed studies compared by valuation perspective, uncertainty, market participation, capacity question and solution method, and the positioning of this study.

Study	Perspective and asset	Uncertainty and information	Markets	Capacity question	Solution method
Helseth et al. (2013)	System; hydrothermal system with pumped storage	Medium to long term; inflow, wind power and prices	System operation (no owner revenue)	Operation of a specified plant	SDDP
Abgottspon and Andersson (2013)	Price-making owner; pumped storage	One year; inflow	DA and forward contracts	Operation of the studied plant	SDP
Abgottspon and Andersson (2014)	Owner; pumped storage	One year; inflow and prices; known or uncertain week compared	Energy and ancillary services	Operation of the studied plant	SDP with stochastic or simplified within-week problems
Helseth et al. (2016)	Price-taking owner; reservoir hydro	Medium term; inflow, energy and reserve-capacity prices	Sequential energy and reserve-capacity markets	Water values and schedules with reserve access	Combined SDP/SDDP
Helseth et al. (2017)	Price-taking owner; reservoir hydro	One year; weekly Markov states	Day-ahead energy and primary reserve capacity	Alternative generator configurations	SDP with weekly MILP
Chazarra et al. (2017)	Price-taking owner; closed-loop pumped storage	Short-term hourly scheduling; known inputs	Energy, secondary reserve and regulation energy	With and without hydraulic short-circuit operation	MILP
Helseth et al. (2018)	System; Nordic power system	One year ahead each week; weather and exogenous prices	System operation (socio-economic surplus)	Transmission and pumped-storage expansions	Two-stage stochastic LP in a rolling horizon
Ak et al. (2019)	Owner; cascade with pumping	Annual evaluation; historical inflows and price years	Energy revenues	Existing cascade versus pumped-storage operation	NLP with operating rules
Dallinger et al. (2019)	System and owner; Austrian plant	One historical year; known inputs	DA and aFRR	Adding pump-turbines	Dispatch model and MILP
Alic et al. (2023)	Owner; pumped storage in a cascade	One year; weekly stages; inflow and energy prices	Price-based energy scheduling	Pumping addition and constraint effects	SDP
Zisos et al. (2023)	Hybrid island-energy design	Long-run simulation; wind, demand and conversion	Supply reliability and economic performance	Reservoir and PV sizing; fixed pump and turbine power	Simulation and optimisation
Schüller et al. (2024)	Price-taking owner; Norwegian watercourse	Inflow and prices	Energy	A new plant with pumping	SDDP-based water values with hourly simulation
Naval and Yusta (2025)	Hybrid-system operator; pumped storage	Hourly operation over a year; wind	Grid exchange and renewable demand coverage	Operation of a specified hybrid configuration	MILP with chance constraints
Bore and Brensholm (2026)	Price-taking owner; reservoir hydro	One year; weekly stages; inflow and prices; known within-week inputs	DA, ID, FCR-N, FCR-D and aFRR capacity; activation excluded	Generating-power upgrades; maximum investment cost	SDDP
This study	**Price-taking owner; pumped storage with natural inflow**	**Weekly stages; inflow and day-ahead prices; known within-week inputs**	**Day-ahead energy and reserve capacity; activation excluded**	**Combinations of pumping power, generating power and reservoir capacity**	**SDDP**

SDP: stochastic dynamic programming; SDDP: stochastic dual dynamic programming; LP: linear programming; MILP: mixed-integer linear programming; NLP: nonlinear programming; PHS: pumped-storage hydropower; DA/ID: day-ahead/intraday; FCR: frequency containment reserve (N: normal operation, D: disturbances); aFRR: automatic frequency restoration reserve. Studies cited only for price behaviour or as general methodological references are not included in the comparison.

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